



CSS (Part 2)

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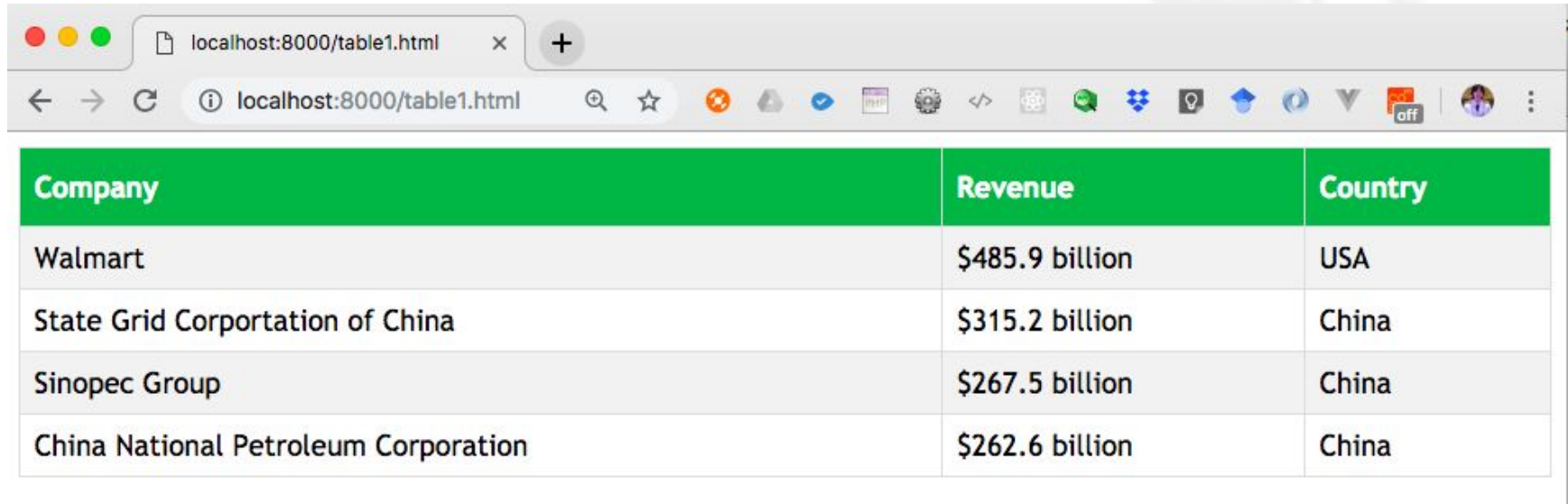


Agenda

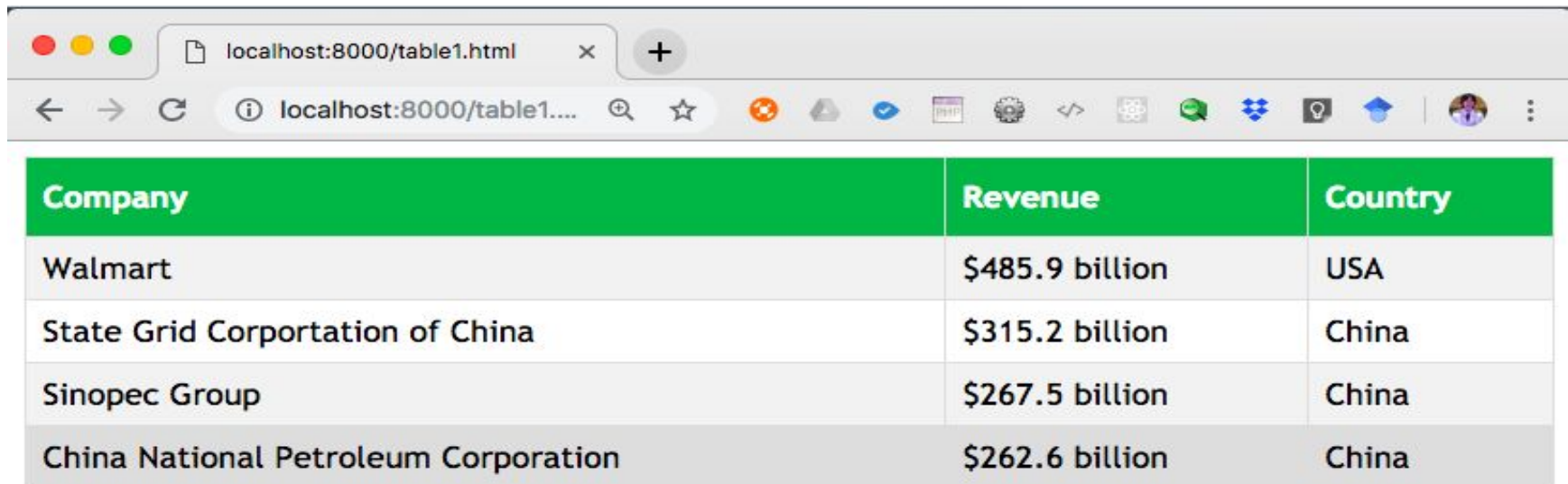
- **CSS Tables**
- CSS Display
- CSS Navigation Bar



CSS Table1 Output



Company	Revenue	Country
Walmart	\$485.9 billion	USA
State Grid Corportation of China	\$315.2 billion	China
Sinopec Group	\$267.5 billion	China
China National Petroleum Corporation	\$262.6 billion	China



Company	Revenue	Country
Walmart	\$485.9 billion	USA
State Grid Corportation of China	\$315.2 billion	China
Sinopec Group	\$267.5 billion	China
China National Petroleum Corporation	\$262.6 billion	China



CSS Table1 Code

```
<style>
#customers {
  border-collapse: collapse;
  width: 100%;
}
#customers td,
#customers th {
  border: 1px solid #ddd;
  padding: 8px;
}
#customers tr:nth-child(even) {
  background-color: #f2f2f2;
}
#customers tr:hover {
  background-color: #ddd;
}
#customers th {
  padding-top: 12px;
  padding-bottom: 12px;
  text-align: left;
  background-color: #4CAF50;
  color: white;
}
</style>
```

```
<table id="customers">
  <tr>
    <th>Company</th>
    <th>Revenue</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Walmart</td>
    <td>$485.9 billion</td>
    <td>USA</td>
  </tr>
  <tr>
    <td>State Grid Corportation of China</td>
    <td>$315.2 billion</td>
    <td>China</td>
  </tr>
  <tr>
    <td>Sinopec Group</td>
    <td>$267.5 billion</td>
    <td>China</td>
  </tr>
  <tr>
    <td>China National Petroleum Corporation</td>
    <td>$262.6 billion</td>
    <td>China</td>
  </tr>
</table>
```



Add a border to a table

← → ↻ ⓘ localhost:8000/table2.html

Add a border to a table:

Firstname	Lastname
Peter	Griffin
Lois	Griffin

Notice that the table in the example above has double borders. This is because both the table and the `<th>` and `<td>` elements have separate borders.

```
<style>
table, th, td {
  border: 1px solid black;
}
</style>
</head>
<body>
<h2>Add a border to a table:</h2>
<table>
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
  </tr>
  <tr>
    <td>Peter</td>
    <td>Griffin</td>
  </tr>
  <tr>
    <td>Lois</td>
    <td>Griffin</td>
  </tr>
</table>
```



Collapse Table Borders

Firstname	Lastname
Peter	Griffin
Lois	Griffin

```
<head>
  <style>
    table {
      border-collapse: collapse;
    }
    table,
    td,
    th {
      border: 1px solid black;
    }
  </style>
</head>
```

```
<body>
  <h2>Let the borders collapse:</h2>
  <table>
    <tr>
      <th>Firstname</th>
      <th>Lastname</th>
    </tr>
    <tr>
      <td>Peter</td>
      <td>Griffin</td>
    </tr>
    <tr>
      <td>Lois</td>
      <td>Griffin</td>
    </tr>
  </table>
```



Horizontal Alignment

- The text-align property sets the horizontal alignment (like left, right, or center) of the content in <th> or <td>
- By default, the content of <th> elements are center-aligned and the content of <td> elements are left-aligned



Horizontal Alignment Example

Firstname	Lastname	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300
Cleveland	Brown	\$250

```
<style>
table, td, th {
    border: 1px solid black;
}
table {
    border-collapse: collapse;
    width: 100%;
}
th {
    text-align: left;
}
</style>
</head>
<body>
```



Vertical Alignment

- The vertical-align property sets the vertical alignment (like top, bottom, or middle) of the content in `<th>` or `<td>`.
- By default, the vertical alignment of the content in a table is middle (for both `<th>` and `<td>` elements).



Vertical Alignment Example

Firstname	Lastname	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300
Cleveland	Brown	\$250

```
<style>
table, td, th {
    border: 1px solid black;
}
table {
    border-collapse: collapse;
    width: 100%;
}
td {
    height: 50px;
    vertical-align: bottom;
}
</style>
```



Table Padding

Firstname	Lastname	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

- `th, td {
padding: 15px;
text-align: left;
}`



Horizontal Dividers

First Name	Last Name	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

```
th, td {  
    border-bottom: 1px solid #ddd;  
}
```



Hoverable Table

- Use the :hover selector on <tr> to highlight table rows on mouse over:

First Name	Last Name	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

- `tr:hover {background-color: #f5f5f5;}`



Striped Tables

First Name	Last Name	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

```
<style>
table {
  border-collapse: collapse;
  width: 100%;
}
th, td {
  text-align: left;
  padding: 8px;
}
tr:nth-child(even) {background-color: #f2f2f2;}
</style>
```



Table Color

First Name	Last Name	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

```
th {  
    background-color: #4CAF50;  
    color: white;  
}
```



Agenda

- CSS Tables
- **CSS Display**
- CSS Navigation Bar



CSS layout: The display Property

- The display property is the most important CSS property for controlling layout
- The display property specifies if/how an element is displayed
- Every HTML element has a default display value depending on what type of element it is
- The default display value for most elements is block or inline



Block-level Elements

The `<div>` element is a block-level element.

- A block-level element always starts on a new line and takes up the full width available
- Examples of block-level elements
 - `<div>`
 - `<h1>` - `<h6>`
 - `<p>`
 - `<form>`
 - `<header>`
 - `<footer>`
 - `<section>`



Inline Elements

This is an inline element inside a paragraph.

- An inline element does not start on a new line and only takes up as much width as necessary
- Examples of inline elements
 -
 - <a>
 -



The display CSS property (1/2)

- The display CSS property is fundamental in controlling how an element is rendered on a web page
- It determines both the display type of an element internally and how it interacts with other elements
- display: block
 - Takes up the full width available
 - Starts on a new line
 - Can have width and height set
 - Examples: <div>, <p>, <h1> to <h6>



The display CSS property (2/2)

- display: inline
 - Takes only as much width as necessary
 - Does not start on a new line
 - Cannot have width and height set
 - Examples: , <a>,
- display: inline-block
 - Combines features of both inline and block
 - Flows with text (inline), but can have width and height (block)
 - Useful for creating horizontal layouts with specific dimensions
- display: none
 - Removes the element from the document flow
 - Element is not rendered and takes up no space



display:none vs visibility:hidden

Use **display: none** — when the element should be gone from the layout, and other content should reflow to fill its space:

1. **Mobile navigation menu (hamburger)** — menu is hidden until clicked; you don't want an empty gap where it would be
2. **Responsive design** — hide a sidebar on small screens so main content takes full width

CSS

```
@media (max-width: 600px) {  
  .sidebar { display: none; }  
}
```

Use **visibility: hidden** — when you want the element invisible but its space preserved, so the layout doesn't shift:

1. **Toggling content without layout jump** — e.g., a "Saved ✓" confirmation that appears/disappears next to a button; keeping its space prevents nearby elements from jumping around
2. **Table columns/rows you may re-show** — keep column widths stable while temporarily hiding data



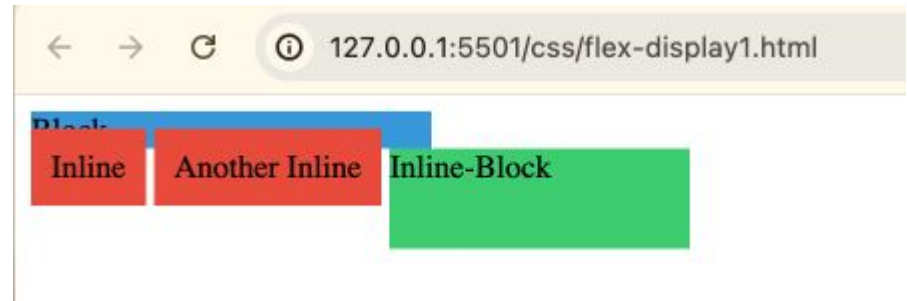
Example of the display CSS property

```
<style>
  .block {
    display: block;
    background-color: #3498db;
    width: 200px;
  }

  .inline {
    display: inline;
    background-color: #e74c3c;
    padding: 10px;
  }

  .inline-block {
    display: inline-block;
    background-color: #2ecc71;
    width: 150px;
    height: 50px;
  }
</style>
```

```
<body>
  <div class="block">Block</div>
  <span class="inline">Inline</span>
  <span class="inline">Another Inline</span>
  <div class="inline-block">Inline-Block</div>
</body>
```



Override the default display value

```
<style>
li {
  display: inline;
}
</style>
</head>
<body>
```

```
<p>Display a list of links as a horizontal menu:</p>
```

```
<ul>
  <li><a href="/html/default.asp" target="_blank">HTML</a></li>
  <li><a href="/css/default.asp" target="_blank">CSS</a></li>
  <li><a href="/js/default.asp" target="_blank">JavaScript</a>
</li>
</ul>
```

Display a list of links as a horizontal menu:

[HTML](#) [CSS](#) [JavaScript](#)



Advanced display values

`display: flex`

- Creates a flex container
- Enables powerful alignment and distribution of space among items

`display: grid`

- Creates a grid container
- Allows for complex two-dimensional layouts

`display: table, table-row, table-cell`

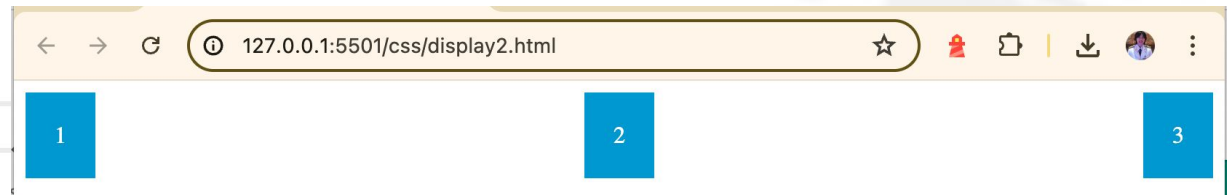
- Used to create table-like layouts without using actual table markup



Example of Using display:flex

```
<head>
  <style>
    .flex-container {
      display: flex;
      justify-content: space-between;
    }

    .flex-item {
      background-color: #3498db;
      padding: 20px;
      color: white;
    }
  </style>
</head>
<body>
  <div class="flex-container">
    <div class="flex-item">1</div>
    <div class="flex-item">2</div>
    <div class="flex-item">3</div>
  </div>
```



flex Property

- Purpose:
 - Used on flex items (children of a flex container)
 - Shorthand for `flex-grow`, `flex-shrink`, and `flex-basis`
- When to use:
 - To control how flex items grow or shrink relative to each other
 - To set the initial main size of a flex item
-



Three flex properties

- flex-grow
 - Determines how much the item will grow relative to other flex items
- flex-shrink
 - Determines how much the item will shrink relative to other flex items
- flex-basis
 - Sets the initial main size of the flex item



flex-grow

- Controls the amount of available space an item should take up
- Default value is 0 (don't grow)
- Higher values make the item grow
- `li.brand { flex-grow: 1;`
/* takes all remaining space; other links pushed right */ }

```
<style>
.container {
  display: flex;
  width: 500px;
}
.item {
  color: white;
  padding: 10px;
  text-align: center;
}
.grow-1 {
  background: red;
  flex-grow: 1;
}
.grow-2 {
  background: green;
  flex-grow: 2;
}
</style>
```



```
<div class="container">
  <div class="item grow-1">1</div>
  <div class="item grow-2">2</div>
  <div class="item grow-1">3</div>
</div>
```

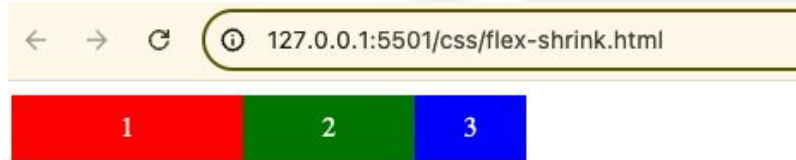


flex-shrink

- Controls how much the item will shrink relative to other items when there isn't enough space
- Default value is 1 (shrink evenly)
- Higher values make the item shrink more

```
<style>
.container {
  display: flex;
  width: 300px;
}
.item {
  color: white;
  padding: 10px;
  text-align: center;
  width: 150px;
}
.shrink-1 {
  background: red;
  flex-shrink: 1;
}
.shrink-2 {
  background: green;
  flex-shrink: 2;
}
.shrink-3 {
  background: blue;
  flex-shrink: 3;
}
</style>
```

```
<div class="container">
  <div class="item shrink-1">1</div>
  <div class="item shrink-2">2</div>
  <div class="item shrink-3">3</div>
</div>
```



- Since the total width of the items = $3 \times 150\text{px} + 3 \times 20\text{px} = 510\text{px}$ and $510\text{px} > 300\text{px}$, the items will shrink to fit the container
- The third item will shrink the most, followed by the second item, and then the first item



flex-basis

- Sets the initial main size of a flex item
- Can be a length (e.g., px, em) or a keyword (e.g., auto)
- Default value is auto
- basis-auto makes the size based on its content
- 0 will be the narrowest
- 100px will have a consistent width regardless of its content

```
<style>
  .container {
    display: flex;
  }
  .item {
    color: white;
    padding: 10px;
    text-align: center;
  }
  .basis-auto {
    background: red;
    flex-basis: auto;
  }
  .basis-0 {
    background: green;
    flex-basis: 0;
  }
  .basis-100 {
    background: blue;
    flex-basis: 100px;
  }
</style>
```

```
<div class="container">
  <div class="item basis-auto">Auto</div>
  <div class="item basis-0">0</div>
  <div class="item basis-100">100px</div>
</div>
```



Agenda

- CSS Tables
- CSS Display
- **CSS Navigation Bar**



Navigation Bars

- Having easy-to-use navigation is important for any web site
- With CSS you can transform boring HTML menus into good-looking navigation bars
- Navigation Bar = List of Links
- A navigation bar needs standard HTML as a base



Navigation1 Example

```
<style>
  ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    width: 200px;
    background-color: #f1f1f1;
  }
  li a {
    display: block;
    color: #000;
    padding: 8px 16px;
    text-decoration: none;
  }
  li a:hover {
    background-color: #555;
    color: white;
  }
</style>
```

← → ↻ ⓘ localhost:8000/navigation1.html#home

Vertical Navigation Bar

Home

News

Contact

About



Pseudo Classes (1/2)

- A pseudo-class is used to define a special state of an element

LVHA order = the required ordering of link pseudo-classes in CSS:

1. **L** — **:link** (unvisited link)
2. **V** — **:visited** (visited link)
3. **H** — **:hover** (mouse over)
4. **A** — **:active** (being clicked)



Pseudo Classes (2/2)

```
a:link { color: blue; }
```

```
a:visited { color: purple; }
```

```
a:hover { background-color: green; } /* must come after :link and :visited */
```

```
a:active { background-color: purple; } /* must come last */
```

- Why order matters: all four selectors have **equal specificity**, so when several apply at once, the *last one in the stylesheet wins* (cascade rule)
- While you're clicking a link, both **:hover** and **:active** are true simultaneously — if **:hover** were written after **:active**, it would override the purple and you'd never see the active color
- Same logic for **:hover** needing to come after **:link/:visited**



Center Links & Add Borders

- Add `text-align:center` to `<li>` or `<a>` to center the links.
- Add the border property to `<ul>` add a border around the navbar.
- If you also want borders inside the navbar, add a `border-bottom` to all `<li>` elements, except for the last one:



Horizontal Navigation Bar

- There are two ways to create a horizontal navigation bar. Using **inline** or **floating** list items.
- Inline List Items
 - One way to build a horizontal navigation bar is to specify the `<li>` elements as inline, in addition to the "standard" code above:

```
<style>
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
}

li {
  display: inline;
}
</style>
</head>
<body>

<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
```



[Home](#) [News](#) [Contact](#) [About](#)



Horizontal Navigation Bar with Flexbox

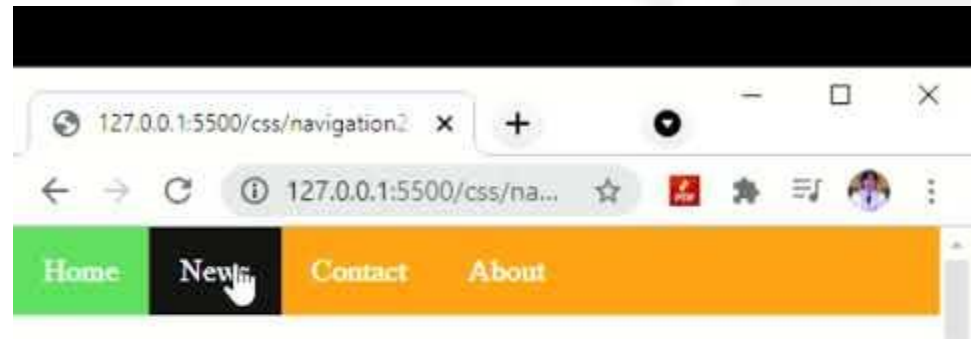
```
ul {  
  display: flex;  
  list-style-type: none; /* no bullet points */  
  margin: 0;           /* no default outer spacing */  
  padding: 0;          /* no default inner spacing */  
  overflow: hidden;  
  background-color: #333;  
}  
li a {  
  display: block;      /* fill clickable area */  
  color: white;  
  text-decoration: none; /* no underline */  
  padding: 14px 16px;  
}
```



Horizontal Navigation Bar Examples

- Create a basic horizontal navigation bar and change the background color of the links when the user moves the mouse over them

```
6 body {
7     margin: 0;
8 }
9 ul {
10     list-style-type: none;
11     margin: 0;
12     padding: 0;
13     overflow: hidden;
14     background-color: #ee8d0e;
15 }
16 li {
17     float: left;
18 }
19 li a {
20     display: block;
21     color: white;
22     text-align: center;
23     padding: 14px 16px;
24     text-decoration: none;
25 }
26
27 li a.active {
28     background-color: #4CAF50;
29 }
30
31 li a:hover {
32     background-color: #111;
33 }
```



Fixed Full-height Side Nav

Try to scroll this area, and see how the sidenav sticks to the page

Notice that this div element has a left margin of 25%. This is because the side navigation is set to 25% width. If you remove the margin, the sidenav

Property overflow:hidden (1/3)

- The overflow property controls what happens to content that is too big to fit into an area
- When you set overflow: hidden on an element, it means
 - Content that exceeds the element's dimensions (height or width) will be clipped and not visible
 - No scrollbars will appear, even if the content overflows
 - The element will still maintain its specified dimensions



Property overflow:hidden (2/3)

- When specifically applied to a `<ul>` element
 - Purpose:
 - It creates a boundary for the list content
 - It prevents list items from visually spilling outside the list container
 - Common use cases
 - Hiding overflowing content in fixed-size containers
 - Controlling layout in responsive design



Property overflow:hidden (3/3)

```
<style>
.container {
  display: flex;
  justify-content: space-around;
}
.list-container {
  width: 200px;
  border: 2px solid #333;
  margin: 20px;
}
h3 {
  text-align: center;
}
ul {
  height: 100px;
  background-color: #f0f0f0;
  padding: 10px;
  margin: 0;
}
.hidden-overflow {
  overflow: hidden;
}
li {
  margin-bottom: 10px;
}

<div class="container">
  <div class="list-container">
    <h3>Without overflow: hidden</h3>
    <ul>
      <li>Item 1</li>
      <li>Item 2</li>
      <li>Item 3</li>
      <li>Item 4</li>
      <li>Item 5</li>
      <li>Item 6</li>
    </ul>
  </div>
  <div class="list-container">
    <h3>With overflow: hidden</h3>
    <ul class="hidden-overflow">
      <li>Item 1</li>
      <li>Item 2</li>
      <li>Item 3</li>
      <li>Item 5</li>
      <li>Item 6</li>
    </ul>
  </div>
</div>
```



Without overflow: hidden

- Item 1
- Item 2
- Item 3
- Item 4
- Item 5
- Item 6

With overflow: hidden

- Item 1
- Item 2
- Item 3
- Item 4



Conclusions

- CSS Tables
 - Styling with borders, alignment, and hover effects
- CSS Display Property
 - Block, inline, inline-block, and flex
- Navigation Bars
 - Creating vertical and horizontal navigation
- Flexbox
 - flex-grow, flex-shrink, and flex-basis properties



References

- W3Schools. CSS Tutorial [Internet]. [cited 2024 Jul 3]. Available from:
<http://www.w3schools.com/css/default.asp>
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